Fresher's Data Science Competition Level 2

IITK Consulting Group

February 2024

CoviStats

Setting

In the fight against the COVID-19 pandemic, the world witnessed lot of vaccines emerging as a saviour for the public. With the promise of developing immunity and suppressing the virus's impact, COVID-19 vaccines became integral to global efforts in overcoming the health crisis. While the availability of these medical resources were critical during this crisis, resource allocation and problem investigation were also important aspects. Many surveys and datas went into generating the best insights.

Well, here again, you are given a dataset with tweets about COVID-19 vaccines and corresponding concern(s) that the tweets are highlighting. Analyze the public's reaction to the vaccines and create a comprehensive pitch giving proper statistical backing to all your hypotheses.

About the Dataset

The file contains `9790` tweets, `6853` labelled with the concerns towards vaccines while `2937` unlabelled.

There are 3 columns in the dataset file:

- `id` : the id of the sample
- `tweet_id` : the id of the tweet
- `content` : the text of the tweet
- `concerns` : vaccine concern(s) expressed in the tweet(separated by spaces).

List of the 12 different `vaccine concerns` in the dataset:

1. `unnecessary` : vaccines are unnecessary, or that alternate cures are better.
2. `mandatory` : Against mandatory vaccination
3. `pharma` : Against Big Pharma
4. `conspiracy` : Some deeper conspiracy (e.g., vaccines are being used to track people, COVID is a hoax)
5. `political` : Political side of vaccines
6. `country`: Country of origin
7. `rushed` : Untested / Rushed Process
8. `ingredients` : Vaccine Ingredients / technology
9. `side-effect` : Side Effects / Deaths
10. `ineffective` : Vaccine is ineffective
11. `religious` : Religious Reasons
12. `none` : No specific reason stated in the tweet, or some reason other than the given ones.

Problem

Analyze and answer the following about this data:

1. Utilizing the training set, develop a model that, given a tweet, predicts the `concerns` that the tweet highlights (It is encouraged to use fundamental techniques instead of deep learning methods).
2. Use the model, trained above, to generate the `concern` labels for the tweets in the test dataset provided. Save it as csv file.
3. Analyse the dataset and for each `concern` label summarise the opinions of the public separately. This means if you have total `n concern` labels, there should ideally be `n` small paragraphs/points/insights. (Hint: *Think about this the NLP way*)
4. Finally, report all your analysis in a presentation. Write all the methods you used even if they didn't yield good results. (*Showcasing your efforts will attract more points*)

Guidelines

"If you can't explain it simply, you don't understand it well enough"

- Albert Einstein

Teams will be asked to present their reports in a presentation held offline.

1. The provided **code** and **statistical report** should be thoroughly annotated and explained.
2. Appropriate **statistical methods** should be employed, with additional points awarded for mentioning the relevance of the chosen methods. **Since you will have to present your work, you should have an understanding of all the methods you are employing. Using code snippets as a blackbox will attract penalties**. Note: Brownie **points for creativity**.
3. It is advisable to work on completing the parts that you find easier, rather than getting stuck on some part.
4. We encourage trying a vast variety of fitting techniques but still having some valid reasoning behind them. **Also don't forget to document your efforts and showcase them**.
5. The **dataset** can be downloaded from here

Timeline

Submission Deadline (For Jupyter Notebooks)	`9/02/2023`
Submission Deadline (For presentation)	`11/02/2023`
Presentation Date and Time	To be announced soon
Results Announcement	After Presentation

Deliverables

- Predictions on test dataset as csv file having the following format
  - `id` : the id of the sample
  - `tweet_id` : the id of the tweet
  - `content` : the text of the tweet
  - `concerns` : vaccine concern(s) expressed in the tweet(separated by spaces).

- Jupyter/Colab Notebook (with the condition that it must give the same results on the test dataset as the submission)
- Presentation which summarizes your efforts as well as the insights for the problems.

Evaluation

Reviews Classification	30%
Statistical Analysis	30%
Codebook and Presentation Submission	10%
Presentation (offline)	30%
